

Example: QR Decomposition Using Gram-Schmidt

Consider the matrix:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Step 1: Extract Columns of A

We define the column vectors:

$$v_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 4 \\ 5 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$$

Our goal is to construct an orthonormal basis q_1, q_2, q_3 .

Step 2: Compute q_1

We normalize v_1 :

$$q_1 = \frac{v_1}{\|v_1\|}$$

$$\|v_1\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$q_1 = \frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{14}} \\ \frac{-1}{\sqrt{14}} \\ \frac{3}{\sqrt{14}} \end{bmatrix}$$

Step 3: Compute q_2

To make v_2 orthogonal to q_1 :

$$u_2 = v_2 - \text{proj}_{q_1}(v_2)$$

where:

$$\text{proj}_{q_1} v_2 = \left(\frac{q_1^T v_2}{q_1^T q_1} \right) q_1$$

Computing $q_1^T v_2$:

$$\begin{aligned} q_1^T v_2 &= \frac{2}{\sqrt{14}}(-1) + \frac{-1}{\sqrt{14}}(4) + \frac{3}{\sqrt{14}}(5) \\ &= \frac{-2 - 4 + 15}{\sqrt{14}} = \frac{9}{\sqrt{14}} \end{aligned}$$

Projection:

$$\begin{aligned}\text{proj}_{q_1} v_2 &= \frac{9}{\sqrt{14}} q_1 = \frac{9}{\sqrt{14}} \begin{bmatrix} \frac{2}{\sqrt{14}} \\ \frac{-1}{\sqrt{14}} \\ \frac{3}{\sqrt{14}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{18}{14} \\ \frac{-9}{14} \\ \frac{27}{14} \end{bmatrix} = \begin{bmatrix} 1.286 \\ -0.643 \\ 1.929 \end{bmatrix}\end{aligned}$$

Now:

$$\begin{aligned}u_2 &= v_2 - \text{proj}_{q_1} v_2 \\ &= \begin{bmatrix} -1 \\ 4 \\ 5 \end{bmatrix} - \begin{bmatrix} 1.286 \\ -0.643 \\ 1.929 \end{bmatrix} \\ &= \begin{bmatrix} -2.286 \\ 4.643 \\ 3.071 \end{bmatrix}\end{aligned}$$

Normalize:

$$\begin{aligned}\|u_2\| &= \sqrt{(-2.286)^2 + (4.643)^2 + (3.071)^2} \\ &= \sqrt{5.23 + 21.57 + 9.43} = \sqrt{36.23} \approx 6.02 \\ q_2 &= \frac{1}{6.02} \begin{bmatrix} -2.286 \\ 4.643 \\ 3.071 \end{bmatrix} = \begin{bmatrix} -0.38 \\ 0.77 \\ 0.51 \end{bmatrix}\end{aligned}$$

Step 4: Compute q_3

Make v_3 orthogonal to both q_1 and q_2 :

$$u_3 = v_3 - \text{proj}_{q_1}(v_3) - \text{proj}_{q_2}(v_3)$$

Following similar computations:

$$q_3 \approx \begin{bmatrix} 0.789 \\ 0.211 \\ 0.577 \end{bmatrix}$$

Step 5: Construct Q and R

The orthogonal matrix Q :

$$Q = \begin{bmatrix} \frac{2}{\sqrt{14}} & -0.38 & 0.789 \\ \frac{-1}{\sqrt{14}} & 0.77 & 0.211 \\ \frac{3}{\sqrt{14}} & 0.51 & 0.577 \end{bmatrix}$$

And the upper triangular matrix R :

$$R = \begin{bmatrix} \sqrt{14} & \frac{9}{\sqrt{14}} & \frac{18}{\sqrt{14}} \\ 0 & 6.02 & 4.77 \\ 0 & 0 & 3.77 \end{bmatrix}$$

Thus, we have:

$$A = QR$$

Final Verification

To verify the decomposition: 1. Check $Q^T Q = I$ (orthonormality). 2. Check $A = QR$ (correctness).